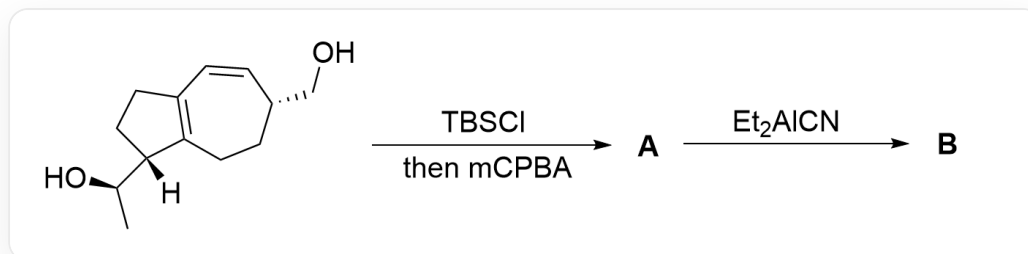


Question



The image describes a two-step organic tandem reaction. The substrate is [H][C@@]1([C@H](O)C)CCC2=C1CC[C@H](CO)C=C2; it reacts with $TBSCl$ followed by $mCPBA$ to obtain **A**; **A** reacts with Et_2AlCN to obtain **B**.

Scientists attempted to synthesize a product with the same molecular skeleton but with a cyano substituent from the synthetic route shown above, but surprisingly, compound **B** without a cyano substituent was produced.

Known:

- B** contains a six-membered ring.
- In the first step, 2.1 equivalents of $TBSCl$ and 1.0 equivalents of $mCPBA$ were added.

Which of the following statements about the structure of **B** is correct:

- A.** All other options are incorrect
- B.** **B** has 5 chiral carbons
- C.** **B** contains 4 oxygen atoms
- D.** **B** Contains fused ring systems
- E.** **B** contains three unsaturated bonds

- F.** **B** first reacts with tetrabutylammonium fluoride, and then undergoes dehydration under acidic conditions. The unsaturation degree of the obtained product is 6.

Answer

Correct Answer: **F**

Detailed Explanation

The substrate contains two hydroxyl groups, which react with 2.1 equivalents of dimethyl tert-butylsilyl chloride, which is an alcohol protection reaction, generating Si – O bonds.

Then it reacts with one equivalent of mCPBA, which is an epoxidation reaction of the double bond; mCPBA preferentially oxidizes the electron-rich tetrasubstituted fused-ring double bond, so the structure of product **A** is [H][C@@]1([C@H](O[Si](C)(C)C(C)(C)C)CCC2(O3)C13CC[C@@H](CO[Si](C)(C)C(C)(C)C)C=C2.

CHECKPOINT

1 PTS

mCPBA preferentially oxidizes the electron-rich tetrasubstituted fused-ring double bond

CHECKPOINT

1 PTS

The structure of **A** is [H][C@@]1([C@H](O[Si](C)(C)C(C)(C)C)CCC2(O3)C13CC[C@@H](CO[Si](C)(C)C(C)(C)C)C=C2

(This question does not require judging the stereochemistry of the epoxide, which is irrelevant to the problem-solving)

Then adding the Lewis acid Et_2AlCN can activate the epoxide; the researcher's original intention was to use cyanide to attack the epoxide to perform ring-opening, but the product does not contain cyanide. At this time, consider that the epoxide is easily ring-opened after being activated, forming a carbocation intermediate; since the

stability of the allylic tertiary carbocation is greater than that of the simple tertiary carbocation, the structure of the carbocation intermediate formed is [H][C@@]1([C@H](O[Si](C)(C)C(C)(C)C)CC[C+]2C1(O[Al](CC)CC)CC[C@@H](CO[Si](C)(C)C(C)(C)C)C=C2.

CHECKPOINT

1 PTS

Lewis acid Et_2AlCN can activate the epoxide; the epoxide is easily ring-opened after being activated, forming a carbocation intermediate

CHECKPOINT

1 PTS

The stability of the allylic tertiary carbocation is greater than that of the simple tertiary carbocation

CHECKPOINT

1 PTS

The structure of the carbocation intermediate is [H][C@@]1([C@H](O[Si](C)(C)C(C)(C)C)CC[C+]2C1(O[Al](CC)CC)CC[C@@H](CO[Si](C)(C)C(C)(C)C)C=C2

At this time, the substrate is a β -carbocation of the hydroxyl group, which is a typical semi-pinacol rearrangement structure. The electron back donation of the hydroxyl group promotes the migration of the carbon atom; because the formation of a six-membered ring is more stable than the formation of a four-membered ring, the structure of the product **B** after migration is O=C1[C@@]([C@H](O[Si](C)(C)C(C)(C)C)([H])CCC12CC[C@@H](CO[Si](C)(C)C(C)(C)C)C=C2.

CHECKPOINT

1 PTS

Semi-pinacol rearrangement occurs; the formation of a six-membered ring is more stable than the formation of a four-membered ring

CHECKPOINT

2 PTS

B is O=C1[C@@]([C@H](O[Si](C)(C)C(C)(C)C)([H])CCC12CC[C@@H](CO[Si](C)(C)C(C)(C)C)C=C2

Determine the options based on the structure of **B**: **B** contains four chiral carbons, two unsaturated bonds, three oxygen atoms, and a spiro system of a five-membered ring and a six-membered ring. Therefore, options B, C, D, and E are all incorrect.

CHECKPOINT

1 PTS

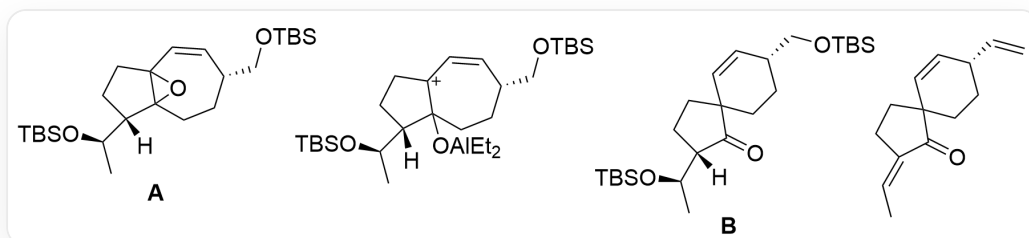
B contains four chiral carbons, two unsaturated bonds, three oxygen atoms, and a spiro system of a five-membered ring and a six-membered ring

B reacts with tetrabutylammonium fluoride to remove all hydroxyl protecting groups, and the alcohol turns into an alkene after heating and dehydration, and the product is O=C1/C(CCC12CC[C@@H](C=C)C=C2)=C\C, which has 6 degrees of unsaturation, so option F is correct.

CHECKPOINT

1 PTS

The reaction product of option F is O=C1/C(CCC12CC[C@@H](C=C)C=C2)=C\C, which has 6 degrees of unsaturation



This figure shows the organic structural formulas involved in this question. The structure of **A** is [H][C@@]1([C@H](O[Si](C)(C)C(C)(C)C)CCC2(O3)C13CC[C@@H](CO[Si](C)(C)C(C)(C)C)C=C; the structure of the carbocation intermediate is [H][C@@]1([C@H](O[Si](C)(C)C(C)(C)C)CC[C+]2C1(O[Al](CC)CC)CC[C@@H](CO[Si](C)(C)C(C)(C)C)C=C2; **B** is O=C1[C@@]([C@H](O[Si](C)(C)C(C)(C)C)C)([H])CCC12CC[C@@H](CO[Si](C)(C)C(C)(C)C)C=C2; the reaction product of option F is O=C1/C(CCC12CC[C@@H](C=C)C=C2)=C\C.